

Calculation Issues

CEO Review Outcome & Tasks

Reviewer: Farhad Abtahi (CEO) · **Source:** "Possible Issues to Confirm" (27 Jul 2026, 20 items) · **Compiled:** 2026-08-03

Farhad reviewed the 19 flagged calculation issues and left 21 comments. This is his verdict on each — what's a real bug to fix, what's confirmed normal/by-design, and what still needs a decision — so we have one agreed reference before the fixes become the next task.

6

CONFIRMED BUGS

8

AGREED AS NORMAL

5

STILL TO DISCUSS

19

ISSUES REVIEWED

1 · Confirmed Bugs — to fix

ID	Issue	CEO decision	Platform
P2	RAMP field / comment / label state three different backward-bending thresholds	"It should be 5° backward. Zero is the main criteria but not practical."	BACKEND
P4	Phone and reports count backward bending differently	"We should match both with -5, same as Web."	MOBILE
P5	Phone warns arm speed at 60°/s, reports at 30°/s	"Phone should become 30 as well."	MOBILE
P6	Mobile PDF prints recommended arm-speed limit of 20°/s	"20 is for Wrist, should be corrected to 30."	MOBILE
P7	Trunk & head bending percentiles never appear in the Word report	"Skip sidebending (not accurate), but we need forward/backward stats for head and Trunk."	BACKEND
P8	Two RAMP colour functions disagree at score 1.5	"It is a bug — should be % of time and will never be 1.5."	BACKEND WEB

⚠ Important nuance on the backward threshold

Backward bending is **-5°** for the **head** (P2/P4, match web). The **trunk-backward risk verdict stays -10°** — Farhad confirmed that one is the deliberate RAMP modification (see P19 below). Don't collapse the two into one value.

Tasks — grouped by who they go to

WEB + BACKEND → WEB DEV

- ☐ **P2 · Fix RAMP head-backward threshold to -5°**
Align the code, the `$\leq -10$` comment, the field name `backward_bending_gt_10_degrees`, and the customer label to -5° . Keep the trunk-backward risk verdict at -10° (intentional).
- ☐ **P7 · Print trunk & head forward/backward bending percentiles in the Word report**
Values are computed but dropped — the Session Percentiles table only prints Elevation Angle + Angular Velocity. Add forward/backward rows for head & trunk. Skip side bending.
- ☐ **P8 · Reconcile the RAMP score-1.5 colour disagreement**
Two colour functions render 1.5 differently. Scoring should be percentage-of-time based and never output 1.5 — fix the backend head-backward ladder and the two web colour functions.
- ☐ **P17 · Add the $^\circ\text{C}$ unit to temperature in the Word report & Web PDF**
Web screen already shows it. **MINOR**

MOBILE → MOBILE DEV

- ☐ **P4 · Match phone backward-bending counting to web (-5°)**
Phone counts any lean past 0° with the report's percentage limits — align both the angle and thresholds to web's -5° .
- ☐ **P5 · Change phone arm-speed warning from $60^\circ/\text{s}$ to $30^\circ/\text{s}$**
- ☐ **P6 · Fix mobile recommended arm-speed limit: $20^\circ/\text{s}$ → $30^\circ/\text{s}$**
The hardcoded 20 is the wrist value; arm should read 30.

2 · Agreed as Normal — by design, not bugs

Confirmed correct or intentional. Recorded so they're not re-flagged later as defects.

ID	Issue	Why it's fine (CEO)
P9	MEC treats the boundary value (8 min/hr) differently across questions	"By design — check the source to verify."
P10	Phone instant feedback ignores backward bending	"By design, backward is too sensitive to give feedback."
P11	Risk is a % of the recording with no minimum length	"We assume it is representative of the whole 8-hour working day."
P13	Web compares median arm speed against the yellow ceiling (30)	"I think it is fine, 30 is a recommended level."
P14	Trunk & head speed is shown but never assessed	"They don't have any recommended limit."
P15	Head side bending is measured but has no risk band	"Deliberate."
P18	Pie charts group arm angles (20/60) differently from the gauges (30/60)	"Different sources (ISO, RAMP, Lund)." — probably fine
P19	"Extension" slice (past -5°) vs "backward" risk row (past -10°)	Both intentional — -5° absorbs sensor noise; -10° is the deliberate RAMP modification.

3 · Still to Discuss — needs a decision

Not settled yet. No code change until these are answered.

ID	Issue	Next step (CEO)
P1	TNO analysis silently discards whole angle ranges	" <i>Might be a bug, but let's skip TNO for now.</i> "
P3	Phone warns about head bending later than the reports (20/45 vs 30)	" <i>Different references (Lund, ISO, RAMP). Let's make a table of the phone and report criteria, then decide.</i> "
P12	Three different rules decide a colour for the same session	" <i>Rule 3 could use a different colour. Make an example report and discuss in group with Liyun and Micke.</i> "
P16	Printed risk-criteria text states ranges that overlap at the edges	" <i>It's fine, but 'less than 5 / more than 10' would be clearer.</i> " — optional wording
P17	Temperature printed with no unit (Word & Web PDF)	" <i>Let's add.</i> " — decided; now a task (see §1)

Source: *Wergonic · Measurement & Risk Calculations* — "*Possible Issues to Confirm*", 27 Jul 2026 (20 items, read against backend staging / mobile main / web dev). Two known-intentional deviations were already excluded from that doc. IDs P1–P19 match the source. P20 was not commented.